# Quizrr Chapter-Wise Test for JEE Main - 2025

**Q.1**

$\int \frac{e^x(1+\sin x)}{1+\cos x}\,\mathrm{d}x$ is equal to

**(A)** $\log|\tan x| + C$

**(B)** $e^x \tan x/2 + C$

**(C)** $e^x \cot x + C$

**(D)** $\sin \log x + C$

**Q.2**

Let $f$ and $g$ be two polynomials then $\int (f(x)g''(x) - f''(x)g(x))\,\mathrm{d}x$ is equal to (ignoring the constant of integration)

**(A)** $\frac{f(x)}{g'(x)}$

**(B)** $f'(x)g(x) - f(x)g'(x)$

**(C)** $f(x)g'(x) - f'(x)g(x)$

**(D)** $f'(x)g'(x) + f'(x)g(x)$

**Q.3**

$\int \frac{\mathrm{d}x}{x(x^4+1)}$ is equal to

**(A)** $\frac{1}{4}\log\frac{x^4+1}{x^4} + C$

**(B)** $\frac{1}{4}\log\left(\frac{x^4}{x^4+1}\right) + C$

**(C)** $\frac{1}{4}\log\left(x^4+1\right) + C$

**(D)** none of these

**Q.4**

If the primitive of $f(x) = \frac{1}{3\sin x + \sin^3 x}$ is equal to $\frac{1}{6}\log\left|\frac{t-1}{t+1}\right| + \frac{1}{12}\log\left|\frac{2+t}{2-t}\right| + C$ then

**(A)** $t = \cos x$

**(B)** $t = \tan x/2$

**(C)** $t = 2\cos x$

**(D)** $t = \sin x$

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**Q.5**

The integral $I = \int 2^{2^x + x} dx = \lambda . 2^{2^x} + C$ (where, $C$ is the constant of integration). Then the value of $\sqrt{\lambda}$ is equal to

**(A)** $\frac{1}{\ln 4}$

**(B)** $\frac{1}{(\ln 2)^2}$

**(C)** $\frac{1}{\ln 2}$

**(D)** $\frac{1}{(\ln 4)^2}$

**Q.6**

$\int (\sin(\log x) + \cos(\log x)) dx =$

**(A)** $x \sin(\log x) + C$

**(B)** $x \cos(\log x) + C$

**(C)** $\sin(\log x) - \cos(\log x) + C$

**(D)** None of these

**Q.7**

$\int x \tan^{-1} x dx =$

**(A)** $\frac{1}{2}\left(x^2 + 1\right) \tan^{-1} x - x$

**(B)** $\frac{1}{2}\left(x^2 + 1\right) \tan^{-1} x - \frac{1}{2}x$

**(C)** $\left(x^2 + 1\right) \tan^{-1} x - x$

**(D)** None of these

**Q.8**

$\int x|x|dx =$

**(A)** $\frac{x^3}{3}$

**(B)** $\frac{x^2|x|}{3}$

**(C)** $\frac{x^2|x|}{2}$

**(D)** None of these

**Q.9**

$\int \frac{dx}{1+\sin x+\cos x} =$

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**(A)** $\log\left|1+\tan\frac{x}{2}\right|+C$

**(B)** $\log\left|1+\sin x+\cos x\right|+C$

**(C)** $2\log\left|1+\tan\frac{x}{2}\right|+C$

**(D)** $\frac{1}{2}\log\left|1+\tan\frac{x}{2}\right|+C$

**Q.10**

$\int\frac{\log x-1}{(\log x)^2}dx=$

**(A)** $\frac{\log x}{x}+C$

**(B)** $\frac{x}{\log x}+C$

**(C)** $\frac{(\log x)^2-x}{\log x}$

**(D)** None of these

**Q.11**

$\int e^x\left(\frac{x-1}{(x+1)^3}\right)dx=$

**(A)** $\frac{e^x}{x+1}$

**(B)** $\frac{e^x}{(x+1)^2}$

**(C)** $\frac{-e^x}{x+1}$

**(D)** $\frac{-e^x}{(x+1)^2}$

**Q.12**

$\int\frac{\sin^8 x-\cos^8 x}{1-2\sin^2 x\cos^2 x}dx$ is equal to

**(A)** $\frac{1}{2}\sin 2x+C$

**(B)** $-\frac{1}{2}\sin 2x+C$

**(C)** $-\frac{1}{2}\sin x+C$

**(D)** $-\sin^2 x+C$

**Q.13**

The value of $\int e^{\tan^{-1}x}\frac{(1+x+x^2)}{1+x^2}dx$ is

**(A)** $xe^{\tan^{-1}x}+C$

**(B)** $\tan^{-1}x+C$

**(C)** $e^{\tan^{-1}x}+2x+C$

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**(D)** None of these

**Q.14**

$\int 32\mathrm{x}^3(\log \mathrm{x})^2\ \mathrm{dx}$ is equal to :

**(A)** $8x^4(\log x)^2 + C$

**(B)** $x^4\left\{8(\log x)^2 - 4(\log x) + 1\right\} + C$

**(C)** $x^4\left\{8(\log x)^2 - 4(\log x)\right\} + C$

**(D)** $x^3\left\{(\log x)^2 - 2\log x\right\} + C$

**Q.15**

If $\int \mathrm{f}(\mathrm{x})\mathrm{dx} = \mathrm{g}(\mathrm{x}) + \mathrm{c}$, then $\int \mathrm{f}^{-1}(\mathrm{x})\mathrm{dx}$ is equal to

**(A)** $xf^{-1}(x) + C$

**(B)** $f\left(g^{-1}(x)\right) + C$

**(C)** $xf^{-1}(x) - g\left(f^{-1}(x)\right) + C$

**(D)** $g^{-1}(x) + C$

**Q.16**

If $\int \frac{xe^x}{\sqrt{1+e^x}}dx = f(x)\sqrt{1+e^x} - 2\log g(x) + C$, then

**(A)** $\mathrm{f}(\mathrm{x}) = \mathrm{x} - 1$

**(B)** $g(x) = \frac{\sqrt{1+\mathrm{e}^x}-1}{\sqrt{1+\mathrm{e}^x}+1}$

**(C)** $g(x) = \frac{\sqrt{1+e^x}+1}{\sqrt{1+e^x}-1}$

**(D)** $f(x) = 2(2-x)$

**Q.17**

What is $\int \sin x \log(\tan x)dx$ equal to?

**(A)** $\cos x \log\tan x + \log\tan(x/2) + c$

**(B)** $-\cos x \log\tan x + \log\tan(x/2) + c$

**(C)** $\cos x \log\tan x + \log\cot(x/2) + c$

**(D)** $-\cos x \log\tan x + \log\cot(x/2) + c$

**Q.18**

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$\int \frac{xe^x}{(1+x)^2} dx =$

**(A)** $-\frac{e^x}{1+x}$

**(B)** $\frac{1+2xe^x}{1+x}$

**(C)** $-\frac{1+2xe^x}{1+x}$

**(D)** $\frac{e^x}{1+x}$

**Q.19**

$\int \frac{x^2}{(x^2+1)(x^2+4)} dx$ is equal to

**(A)** $\tan^{-1} x + 2\tan^{-1}\left(\frac{x}{2}\right) + C$

**(B)** $\tan^{-1}\left(\frac{x}{2}\right) - 4\tan^{-1} x + C$

**(C)** $-\frac{1}{3}\tan^{-1} x + \frac{2}{3}\tan^{-1}\left(\frac{x}{2}\right) + C$

**(D)** $4\tan^{-1}\left(\frac{x}{2}\right) - 2\tan^{-1} x + C$

**Q.20**

$\int \frac{(1+x)}{x(1+xe^x)^2} dx$

**(A)** $\ln\left|\frac{xe^x}{1+xe^x}\right| + \frac{1}{1+xe^x} + C$

**(B)** $(1+xe^x) + \ln\left|\frac{xe^x}{1+xe^x}\right| + C$

**(C)** $\frac{1}{1+xe^x} + \ln\left|xe^x(1+xe^x)\right| + C$

**(D)** None of these

## Answers & Solutions

**Q.1 Answer:**

$e^x \tan x/2 + C$

**Solution:**

$\int \frac{e^x(1+\sin x)}{1+\cos x}\,\mathrm{d}x$
$= \int \frac{e^x[1+2\sin(x/2)\cos(x/2)]}{2\cos^2(x/2)}\mathrm{d}x$
$= \int e^x\left(\frac{1}{2}\sec^2\frac{x}{2} + \tan\frac{x}{2}\right)\mathrm{d}x$
$= \frac{1}{2}\int e^x \sec^2\frac{x}{2}\,\mathrm{d}x + \int e^x \tan\frac{x}{2}\,\mathrm{d}x$
$= \frac{1}{2}\int e^x \sec^2\frac{x}{2}\,\mathrm{d}x + e^x\tan\frac{x}{2} - \frac{1}{2}\int e^x\sec^2\frac{x}{2} + C$

**[integrating by parts]**

$= e^x \tan\frac{x}{2} + C$

**Q.2 Answer:**

$f(x)g'(x) - f'(x)g(x)$

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**Solution:**

$$\begin{aligned}&\int (f(x)g''(x) - f''(x)g(x))\,\mathrm{d}x\\ &= \quad f(x)g'(x) - \int f'(x)g'(x)\mathrm{d}x\\ &\quad - \left[g(x)f(x) - \int f'(x)g'(x)\mathrm{d}x\right]\\ &= f(x)g'(x) - g(x)f'(x)\end{aligned}$$

**Q.3 Answer:**

$\frac{1}{4}\log\left(\frac{x^4}{x^4+1}\right) + C$

**Solution:**

$$\begin{aligned}&\int \frac{\mathrm{d}x}{x(x^4+1)} = \frac{1}{4}\int \frac{4x^3\,\mathrm{d}x}{x^4(x^4+1)}\\ &= \frac{1}{4}\int \frac{\mathrm{d}t}{t(t+1)} \quad (t = x^4)\\ &= \frac{1}{4}\int \left(\frac{1}{t} - \frac{1}{t+1}\right)\mathrm{d}t = \frac{1}{4}\log\frac{x^4}{x^4+1} + C\end{aligned}$$

**Q.4 Answer:**

$t = \cos x$

**Solution:**

$$\begin{aligned}&\int \frac{\mathrm{d}x}{3\sin x + \sin^3 x} = \int \frac{\sin x\,\mathrm{d}x}{(3+\sin^2 x)\sin^2 x}\\ &= \int \frac{\sin x\,\mathrm{d}x}{(4-\cos^2 x)(1-\cos^2 x)}\\ &= -\int \frac{\mathrm{d}t}{(4-t^2)(1-t^2)}\\ &= -\frac{1}{3}\int \left(\frac{1}{t^2-4} - \frac{1}{t^2-1}\right) \quad (t = \cos x)\\ &= \frac{1}{6}\log\left|\frac{t-1}{t+1}\right| + \frac{1}{12}\log\left|\frac{2+t}{2-t}\right| + C\end{aligned}$$

**Q.5 Answer:**

$\frac{1}{\ln 2}$

**Solution:**

**Given,** $I = \int 2^{2^x}.2^x dx$

**Let,** $2^{2^x} = t \Rightarrow 2^{2^x}.2^x.\ln 2^2 dx = dt$

$\Rightarrow 2^{2^x}.2^x dx = \frac{dt}{\ln 2^2}$

**Thus,** $I = \int \frac{dt}{\ln 2^2} = \frac{t}{\ln 2^2} + C$

$= \frac{2^{2^x}}{\ln 2^2} + C$

$\therefore \lambda = \frac{1}{\ln 2^2} \Rightarrow \sqrt{\lambda} = \frac{1}{\ln 2}$

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**Q.6 Answer:**

$x \sin(\log x) + C$

**Solution:**

$I = \int \sin(\log x) \cdot 1 dx + \int \cos(\log x) \cdot 1 dx$

**(Integrating by parts the first integral)**

$= \sin(\log x) \cdot x - \int \cos(\log x) \cdot \frac{1}{x} x dx + \int \cos(\log x) \cdot dx$

$= x \sin(\log x) + C$

**Q.7 Answer:**

$\frac{1}{2}\left(\mathrm{x}^2+1\right) \tan^{-1} \mathrm{x}-\frac{1}{2} \mathrm{x}$

**Solution:**

$I = \left(\tan^{-1} x\right) \frac{x^2}{2} - \int \frac{1}{1+x^2} \cdot \frac{x^2}{2} dx$

$= \frac{1}{2} x^2 \tan^{-1} x - \frac{1}{2} \int \frac{1+x^2-1}{1+x^2} dx$

$= \frac{1}{2} \mathrm{x}^2 \tan^{-1} \mathbf{x} - \frac{1}{2} \int \left(1 - \frac{1}{1+\mathrm{x}^2}\right) \mathrm{dx}$ **etc.**

**Q.8 Answer:**

$\frac{\mathbf{x}^2|\mathrm{x}|}{3}$

**Solution:**

$\mathrm{I} = \int \mathrm{x}.|\mathrm{x}| \mathrm{dx} = \int |\mathrm{x}| \cdot \mathrm{xdx}$

**(Integrating by parts taking $|\mathbf{x}|$ as first function)**

$= |x| \cdot \frac{x^2}{2} - \int \frac{|x|}{x} \cdot \frac{x^2}{2} dx = \frac{x^2|x|}{2} - \frac{1}{2} \int |x| x dx$

$\Rightarrow \mathrm{I}\left(1+\frac{1}{2}\right) = \frac{\mathrm{x}^2|\mathrm{x}|}{2} \Rightarrow \mathrm{I} = \frac{1}{3} \mathrm{x}^2|\mathrm{x}|$

**Q.9 Answer:**

$\log \left|1 + \tan \frac{x}{2}\right| + C$

**Solution:**

**We have,** $1 + \sin x + \cos x = (1 + \cos x) + \sin x$

$= 2 \cos^2 \left(\frac{x}{2}\right) + 2 \sin \left(\frac{x}{2}\right) \cos \left(\frac{x}{2}\right),$

$= 2 \cos^2 \left(\frac{x}{2}\right) \left\{1 + \frac{\sin\left(\frac{x}{2}\right)}{\cos\left(\frac{x}{2}\right)}\right\}$

$= \frac{2}{\sec^2\left(\frac{x}{2}\right)} \left\{1 + \tan \left(\frac{x}{2}\right)\right\}$

$\therefore I = \int \frac{1}{1+\sin x+\cos x} dx$

$= \int \frac{\frac{1}{2} \cdot \sec^2\left(\frac{x}{2}\right)}{1+\tan\left(\frac{x}{2}\right)} dx$

$= \int \frac{\frac{d}{dx}\left(1+\tan\left(\frac{x}{2}\right)\right)}{1+\tan\left(\frac{x}{2}\right)} dx$

$= \ln \left|1 + \tan \left(\frac{x}{2}\right)\right| + C,$

**Q.10 Answer:**

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$\frac{x}{\log x} + C$

**Solution:**

$I = \int \frac{1}{\log x} - \frac{1}{(\log x)^2} dx = \int \frac{1}{\log x} dx - \int \frac{1}{(\log x)^2} dx$

**(Integrating by parts the first integral)**

$= \frac{1}{\log x} \cdot x - \int \frac{-1}{(\log x)^2} \cdot \frac{1}{x} \cdot x dx - \int \frac{1}{(\log x)^2} dx$

$= \frac{\mathrm{x}}{\log \mathrm{x}} + \mathrm{C}$

**Q.11 Answer:**

$\frac{e^x}{(x+1)^2}$

**Solution:**

Let $I = \int \frac{(x-1)e^x}{(x+1)^3} dx$

$\Rightarrow I = \int \left\{ \frac{x+1-2}{(x+1)^3} \right\} e^x dx$

$= \int \left\{ \frac{1}{(x+1)^2} - \frac{2}{(x+1)^3} \right\} e^x dx$

$= \int e^x \cdot \frac{1}{(x+1)^2} dx - 2 \int e^x \frac{1}{(x+1)^3} dx$

**Applying integrating by parts, we get**

$= \left( \frac{1}{(x+1)^2} e^x - \int e^x \frac{(-2)}{(x+1)^3} dx \right) -$

$2 \int e^x \frac{1}{(x+1)^3} dx$

$= \frac{e^x}{(x+1)^2} + C$

**Q.12 Answer:**

$-\frac{1}{2} \sin 2x + C$

**Solution:**

$\frac{\sin^8 x - \cos^8 x}{1 - 2\sin^2 x \cos^2 x} = \frac{\left(\sin^4 x + \cos^4 x\right)\left(\sin^4 x - \cos^4 x\right)}{\left(\sin^2 x + \cos^2 x\right)^2 - 2\sin^2 x \cos^2 x} =$

$\frac{\left(\sin^4 x + \cos^4 x\right)\left(\sin^4 x - \cos^4 x\right)}{\sin^4 x + \cos^4 x}$

$= \left(\sin^2 x + \cos^2 x\right)\left(\sin^2 x - \cos^2 x\right) = -\cos 2x$

$\int \frac{\sin^8 x - \cos^8 x}{1 - 2\sin^2 x \cos^2 x} dx = -\int \cos 2x dx = -\frac{1}{2} \sin 2x + c$

**Q.13 Answer:**

$xe^{\tan^{-1} x} + C$

**Solution:**

**Put** $\mathrm{x} = \tan\theta \Rightarrow \mathrm{dx} = \sec^2\theta \mathrm{d}\theta$

$\mathrm{I} = \int \mathrm{e}^{\theta} \frac{1 + \tan\theta + \tan^2\theta}{1 + \tan^2\theta} \cdot \sec^2\theta \mathrm{d}\theta$

$= \int \mathrm{e}^{\theta} \left(\tan\theta + \sec^2\theta\right) \mathrm{d}\theta$

$= \mathrm{e}^{\theta} \tan\theta + \mathrm{c} = xe^{\tan^{-1} x} + C$

**Q.14 Answer:**

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$x^4\left\{8(\log x)^2-4(\log x)+1\right\}+C$

**Solution:**

Let $I=\int 32x^3(\log x)^2dx$

$=32\left\{(\log x)^2\frac{x^4}{4}-\int 2\log x\frac{1}{x}\cdot\frac{x^4}{4}dx\right\}$

$=\frac{32}{4}x^4(\log x)^2-16\int x^3\log xdx$

$=8x^4(\log x)^2-4x^4\log x+4\int x^3dx$

$=x^4\left\{8(\log x)^2-4\log x+1\right\}+C$

**Q.15 Answer:**

$xf^{-1}(x)-g\left(f^{-1}(x)\right)+C$

**Solution:**

Let $\mathrm{I}=\int \mathrm{f}^{-1}(\mathrm{x})\mathrm{dx}$

and $\mathrm{f}^{-1}(\mathrm{x})=\mathrm{t}\Rightarrow \mathrm{x}=\mathrm{f}(\mathrm{t})\Rightarrow \mathrm{dx}=\mathrm{f}'(\mathrm{t})\mathrm{dt}$

Put value of $\mathrm{dx}$ and $\mathrm{f}^{-1}(\mathrm{x})$ in I, we get $I=\int tf'(t)dt$

Now, integrate it by parts, $I=tf(t)-\int f(t)dt$

Given, $\int f(x)dx=g(x)+C$

$\therefore \mathrm{I}=\mathrm{tf}(\mathrm{t})-[\mathrm{g}(\mathrm{t})]+\mathrm{C}$

Now, by putting value of $\mathrm{t},\mathrm{f}(\mathrm{t})$ and $\mathrm{g}(\mathrm{t})$ we get, $\mathrm{I}=\mathrm{xf}^{-1}(\mathrm{x})-\mathrm{g}\left[\mathrm{f}^{-1}(\mathrm{x})\right]+\mathrm{C}$

**Q.16 Answer:**

$g(x)=\frac{\sqrt{1+\mathrm{e}^x}-1}{\sqrt{1+\mathrm{e}^{\mathrm{x}}}+1}$

**Solution:**

$I=\int\frac{xe^x}{\sqrt{1+e^x}}dx$ we have

$\int\frac{\mathrm{e}^{\mathrm{x}}}{\sqrt{1+\mathrm{e}^{\mathrm{x}}}}\mathrm{dx}=2\sqrt{1+\mathrm{e}^{\mathrm{x}}}$

Integrating I by parts with x as first function $I=x.2\sqrt{1+e^x}-\int 2\sqrt{1+e^x}dx$

$=2\mathrm{x}\sqrt{1+\mathrm{e}^{\mathrm{x}}}-2\int \mathrm{t}\cdot\frac{2\mathrm{tdt}}{\mathrm{t}^2-1}$ ( Putting $1+\mathrm{e}^{\mathrm{x}}=\mathrm{t}^2$)

$=2x\sqrt{1+e^x}-4\int\frac{t^2-1+1}{t^2-1}dt$

$=2x\sqrt{1+e^x}-4\left[t+\frac{1}{2}\log\frac{t-1}{t+1}\right]+c$

$=2(x-2)\sqrt{1+e^x}-2\log\left(\frac{\sqrt{1+e^x}-1}{\sqrt{1+e^x}+1}\right)+c$

**Q.17 Answer:**

$-\cos x\log\tan x+\log\tan(x/2)+c$

**Solution:**

$\int\sin x\log(\tan x)dx$

$=-\cos x\log\tan x-\int(-\cos x)\frac{1}{\tan x}\cdot\sec^2 xdx$

$=-\cos x\log\tan x+\int\frac{1}{\sin x}dx$

$=-\cos x\log(\tan x)+\int\frac{1+\tan^2\frac{x}{2}}{2\tan\frac{x}{2}}dx$

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**Now putting $\frac{x}{2} = t$, we get,**

$= -\cos x \log \tan x + \int \frac{1}{t} \cdot dt$

$= -\cos x \log \tan x + \log(t) + c$

$= -\cos x \log \tan x + \log \tan \left(\frac{x}{2}\right) + c$

**Q.18 Answer:**

$\frac{e^x}{1+x}$

**Solution:**

$\int \frac{xe^x}{(1+x)^2} dx = \int \left(\frac{x+1-1}{(1+x)^2}\right)^x dx$

$= \int \left(\frac{1}{1+x} - \frac{1}{(1+x)^2}\right) e^x dx = \frac{e^x}{1+x} + C$

**Q.19 Answer:**

$-\frac{1}{3}\tan^{-1} x + \frac{2}{3}\tan^{-1}\left(\frac{x}{2}\right) + C$

**Solution:**

$\int \frac{x^2}{(x^2+1)(x^2+4)} dx.$

$= \frac{1}{3}\int \left[\frac{4}{x^2+4} - \frac{1}{x^2+1}\right] dx$

$= -\frac{1}{3}\int \frac{1}{x^2+1} dx + \frac{4}{3}\int \frac{1}{x^2+4} dx$

$= -\frac{1}{3}\tan^{-1} x + \frac{4}{3} \times \frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right) + C$

$= -\frac{1}{3}\tan^{-1} x + \frac{2}{3}\tan^{-1}\left(\frac{x}{2}\right) + C$

**Q.20 Answer:**

$\ln\left|\frac{xe^x}{1+xe^x}\right| + \frac{1}{1+xe^x} + C$

**Solution:**

$\mathrm{I} = \int \frac{(1+\mathrm{x})}{\mathrm{x}(1+\mathrm{xe}^{\mathrm{x}})^2} \mathrm{dx} = \int \frac{(1+\mathrm{x})\mathrm{e}^{\mathrm{x}}}{\mathrm{xe}^{\mathrm{x}}(1+\mathrm{xe}^{\mathrm{x}})^2} \mathrm{dx}$

**Put** $xe^x = t \Rightarrow (e^x + xe^x)\, dx = dt$

$\mathrm{I} = \int \frac{\mathrm{dt}}{\mathrm{t}(1+\mathrm{t})^2}$

**Let** $\frac{1}{t(1+t)^2} = \frac{A}{t} + \frac{B}{1+t} + \frac{D}{(1+t)^2}$**, we get**

$A = \frac{1}{(1+0)^2} = 1, D = \frac{1}{-1} = -1$

**Equating coefficient of** $t^2, 0 = A + B \Rightarrow B = -1$

$\therefore \mathrm{I} = \int \left[\frac{1}{t} - \frac{1}{1+\mathrm{t}} - \frac{1}{(1+\mathrm{t})^2}\right] \mathrm{dt}$

$= \ln|t| - \ln|1+t| + \frac{1}{1+t} + C$

$= \ln\left|\frac{t}{1+t}\right| + \frac{1}{1+t} + C$

$= \ln\left|\frac{xe^x}{1+xe^x}\right| + \frac{1}{1+xe^x} + C$

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